Implementation and Study of the Dieckmann-Law Population Dynamics Model: Numerical Analysis and Comparison with the Neighborhood-Dependent Model

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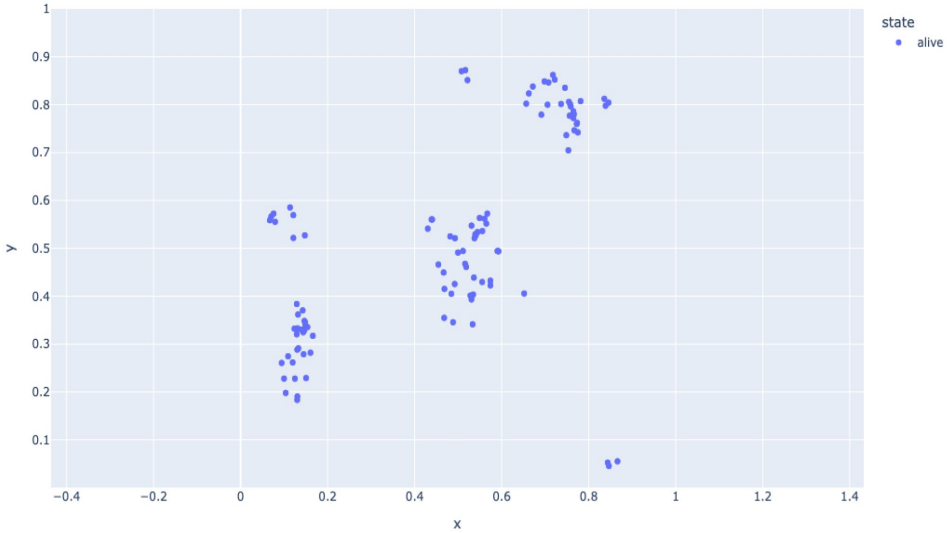
Introduction

Introduction

- Population dynamics is the change in the size and structure of a population.
- An N×N spatial domain.
- Individuals reproduce and die.



Models

Neighborhood-Dependent Model:

Birth Rate:

$$N_R^{(i)} = \sum \mathbf{1}_{\{(x_i - x_j)^2 + (y_i - y_j)^2 < R^2\}}$$

$$\lambda_i = \max\left(0,\ \lambda_0 - \frac{1}{N_s} N_R^{(i)}\right) \quad P(\text{birth}_i) = \lambda_i \Delta t$$

Death Rate:

$$P(\text{death}_i) = \mu_i \Delta t = \mu_0 \Delta t$$

Individuals move randomly; the offspring appears at the same location as the parent.

Dieckmann-Law Model:

Birth Rate:

$$P(\text{birth}_i) = \lambda_0 \Delta t.$$

Death Rate:

$$C_i = \sum_{j \neq i} \exp\left(-\frac{(x_i - x_j)^2 + (y_i - y_j)^2}{2R_c^2}\right)$$

$$\mu_i = \mu_0 + \alpha C_i \qquad P(\text{death}_i) = \min(1, \mu_i \Delta t)$$

Individuals do NOT move; the offspring coordinates are determined by normal noise.

Program Implementation

Implementation

- Simulation in C++.
- Visualization in Python (библиотека plotly)
- Data storage via CSV files.

Asymptotic Complexity

Neighborhood-Dependent Model:

$$T_{\mathrm{ND}}(N) = O(N^2)$$

Dieckmann-Law Model:

$$T_{\mathrm{DL}}(N) = O(N^2)$$

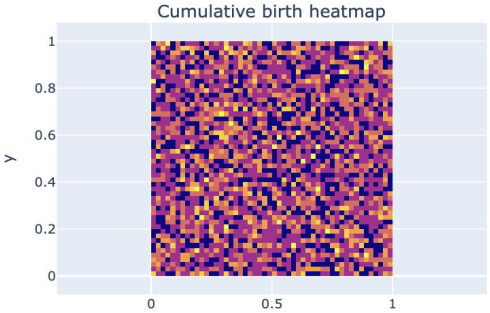
Exponential growth at high birth rates!

Numerical Analysis of the Dieckmann-Law Model

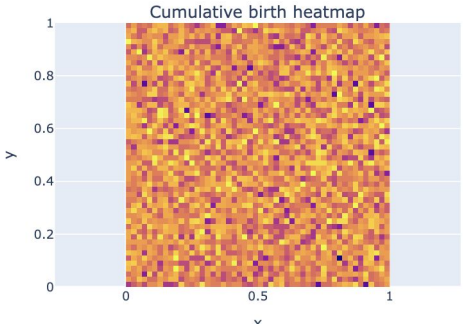
Series 1: Interaction Radius and Offspring Spread

Birth heatmap over the field for different parameter values:

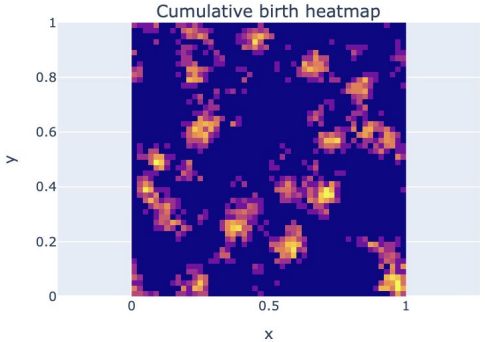
Rd = 0.12
Rc = 0.12



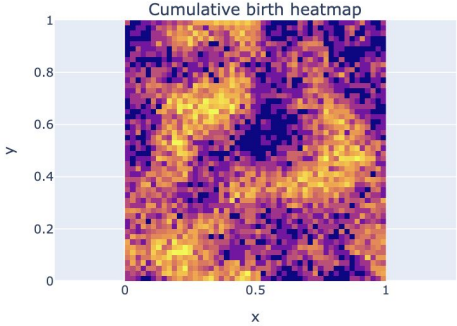
Rd = 0.12
Rc = 0.02



Rd = 0.02
Rc = 0.12



Rd = 0.02
Rc = 0.02

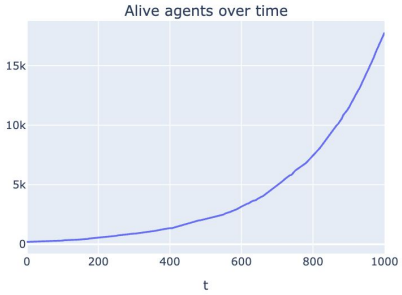


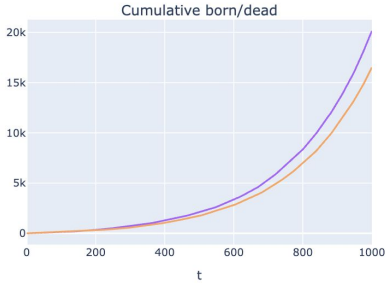
Series 2: Kernel Coefficient

Let us consider the clustering regime (Rd = 0.02, Rc = 0.12).

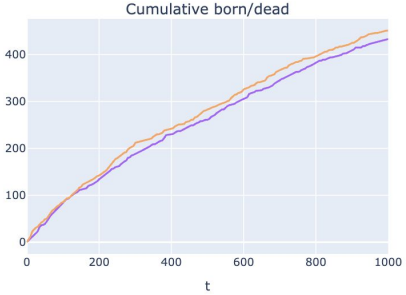
comp_stg = 0.00258 (critical value) !

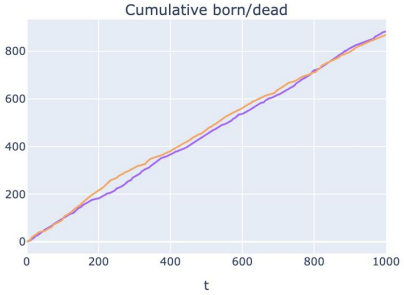
comp_stg = 1e-6 (weak comp.)





comp_stg = 0.0030 (strong comp.)





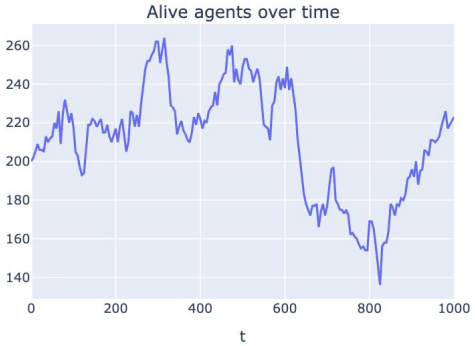
Series 3: Birth Rate / Death Rate

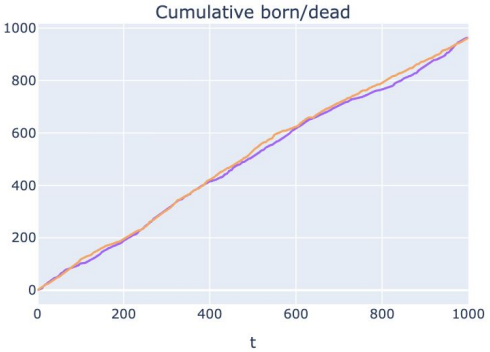
Let us consider the clustering regime (Rd = 0.02, Rc = 0.12). Even when the birth rate exceeds the death rate, extinction is possible (due to clustering).

b0 = 0.22
d0 = 0.20

b0 = 0.20
d0 = 0.20

b0 - d0 = 0.042 (critical value) !



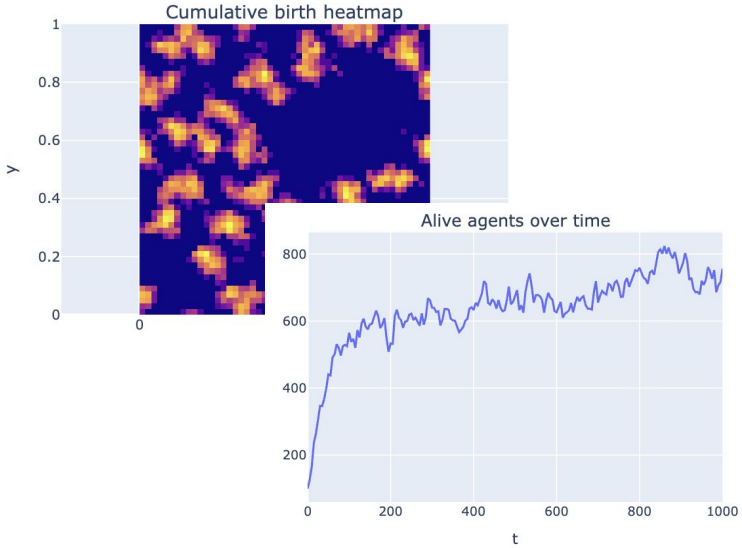


Comparison with the Neighborhood-Dependent Model

Series 1: Structural Differences

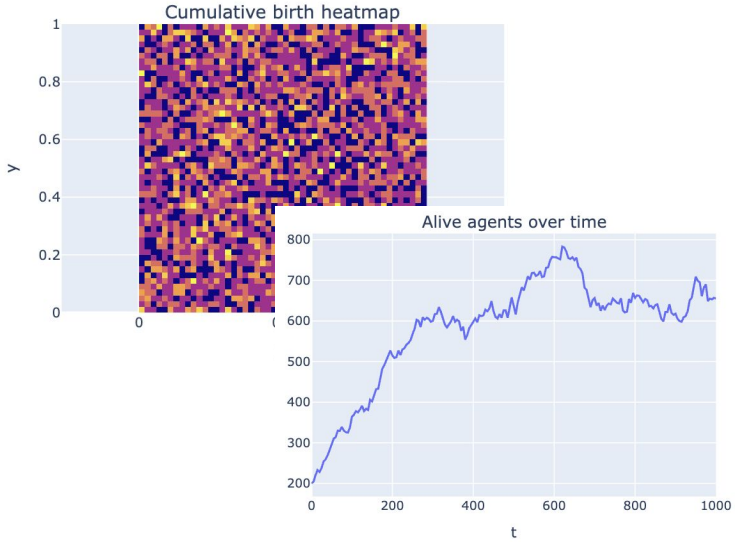
Neighborhood-Dependent Model:

Birth Heatmap and Dynamics:



Dieckmann-Law Model:

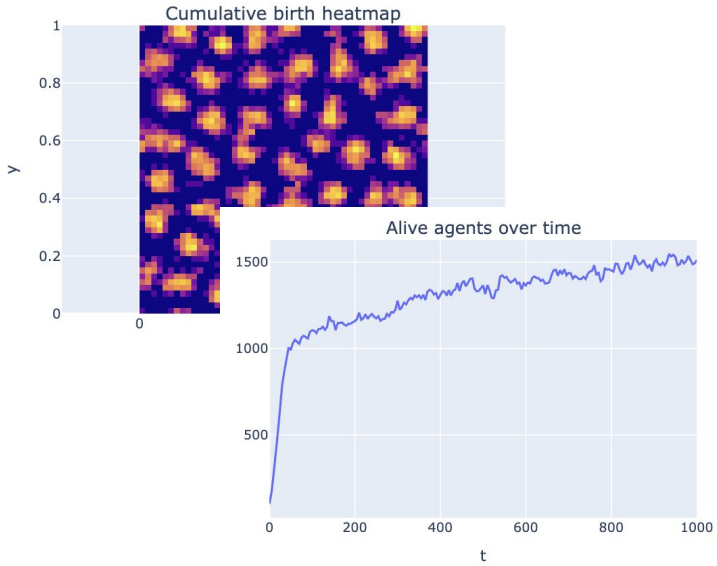
Birth Heatmap and Dynamics:



Series 2: Behavior under Clustering

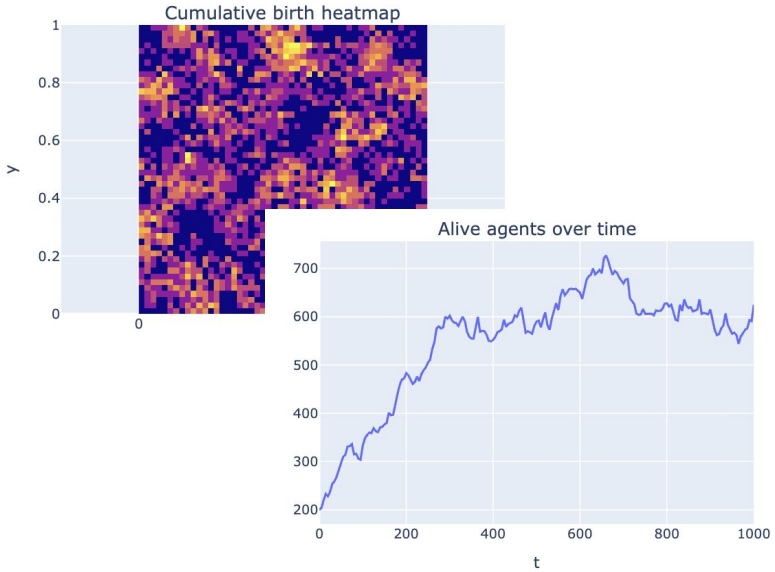
Neighborhood-Dependent Model:

Birth Heatmap and Dynamics::



Dieckmann-Law Model:

Birth Heatmap and Dynamics::



Series 3: Competition Mechanisms

Neighborhood-Dependent Model:

Heatmap and Dynamics: (R >>):

Dieckmann-Law Model:

Dynamics (Rd >>, Rc <<):

Series 4: Behavior under Extinction

Neighborhood-Dependent Model:

Unstable fluctuations.

Dieckmann-Law Model:

Smooth decline.